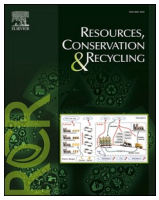
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Full length article



# The water use of data center workloads: A review and assessment of key determinants

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ABSTRACT

The global importance of data center water use is increasing with the rapid growth of digitalization and artificial intelligence. This study analyzes the factors influencing workload-level water use, measured in liters consumed per workload, to guide water-saving strategies in data centers. Our findings reveal workload-level water use variations exceeding 10,000-fold, driven by over 1000-fold differences in water consumption per kilowatt hour of server electricity consumed and approximately 10-fold differences in server workload efficiency. Key determinants are ranked as server efficiency, electrical grid water consumption factors, server utilization, cooling system type, infrastructure efficiency, climate zone, inactive server percentage, and server refresh cycle. Notably, there is no single recipe for minimizing water use; instead, optimal outcomes depend on tailored combinations of these factors. This analysis addresses critical knowledge gaps by identifying the determinants of data center water use and exploring their achievable minima under diverse site-specific constraints.

## 1. Introduction

Increasing demand for digital services has led to an increased reliance on data centers—the backbone of the internet which provides the necessary functions of data processing, communication, and storage (Shehabi et al., 2024, 2016). However, the operation of data centers requires significant amounts of electricity and water (Masanet et al., 2020; Mytton, 2021). While data center electricity use has been the subject of much investigation (Andrae and Edler, 2015; Belkhir and Elmeligi, 2018; Jones, 2018; N. Lei and Masanet, 2021; Masanet et el., 2020), data center water consumption is a less studied topic that is becoming increasingly important in the context of water scarcity and freshwater conservation, especially at local levels (Lei and Masanet, 2022; Mytton, 2021; M.A.B. Siddik et al., 2021). Many data centers can consume a significant amount of water directly for cooling and space conditioning (Lei and Masanet, 2022). For example, Google's data centers in The Dalles consumed 355 million gallons of water in 2021, representing 29 % of the city's total water consumption, which has raised concerns about local water stress, particularly in light of the recent drought conditions in this region (Oregon Tech, 2022). Additionally, data centers consume water indirectly through the water used for electricity generation (Mytton 2021; Shehabi et al., 2016; Siddik et al., 2024, M.A.B. 2021). This type of water use can be significant at the local or national level depending on the power generation technologies, power mixes, and electricity usage of the data centers. For example, some very large data centers have power capacities of hundreds of megawatts and the generation of 1 kW-hour of thermoelectricity or hydropower could consume 0.18–2.0 liters or 0.67–1194 liters of water (Lee et al., 2018), respectively. This demonstrates a significant amount of data center indirect water use and further emphasizes the importance of considering data center water use going forward.

To align with the growing importance of water conservation and sustainability in data centers, operators are beginning to track, manage, and report their water usage effectiveness (WUE) at the site level (WUE-site) and at the source level (WUE-source) (Facebook, 2014; Scaleway, 2021). WUE-site measures the efficiency of onsite water consumption at a data center facility, expressed as the ratio of the data center's total onsite water use to its information technology (IT) equipment electricity use. WUE-source quantifies water use efficiency at the source level by considering the total water use (including on-site and off-site water) in relation to IT equipment electricity use (The Green Grid 2011a). WUE-site and WUE-source are becoming more commonly-reported

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metrics within the data center industry due to their simplicity (Meta Sustainability, 2023). However, much like power utilization effectiveness (PUE) (Masanet et al., 2013), they do not consider the hardware efficiency or computing performance of a data center (Horner and Azevedo, 2016). Therefore, WUE-site and WUE-source cannot be used to compare the water use of data centers with different workload quantities and efficiencies.

In addition, systematic reporting and research related to data center water uses and their determinants remain limited in the literature, which poses challenges for designers, engineers, operators, investors, and policymakers in effectively understanding and addressing data center water usage issues.

- First, while reporting water use and efficiency metrics such as PUE and WUE-site is becoming more common in the industry, these metrics are primarily reported by cloud providers, with significant variation in scope, quality, and granularity (Masanet et al., 2024). Companies like Google, Meta, Microsoft, and Scaleway serve as role models for reporting standards (Google, 2024; Meta Sustainability, 2023; Microsoft, 2024; Scaleway, 2021), yet even their reports lack sufficient detail on water use determinants needed to inform effective water-saving strategies.
- Second, limited quantitative studies have explored the impact of cooling systems and economizers on WUE-site (Gozcu et al., 2017; Lei and Masanet, 2022; Sharma et al., 2009, 2008). Among these, (Lei and Masanet, 2022) offers the most comprehensive analysis, incorporating data center operating efficiency, climate conditions, and the combined effects of cooling systems and economizers on both PUE and WUE-site. However, this study does not account for the water that is indirectly used in electricity generation. Conversely, research from AB Siddik (M.A.B. Siddik et al., 2021) includes indirect water consumption but relies on summarized PUE statistics by data center size, derived from Masanet et al. (Masanet et el., 2020), which limits the ability to identify specific determinants of water use and develop targeted saving strategies.
- Third, another line of research conducted by Lawrence Berkeley National Laboratory has focused on estimating data center water use at the regional level (Shehabi et al., 2024, 2016). Their analysis estimated that U.S. data centers directly consumed approximately 21.2 billion liters of water in 2014 and 66 billion liters in 2023. However, these studies disaggregate water use by space type and lack fine-grained analyses of the factors influencing water use, hindering the identification of tailored strategies to improve water efficiency.
- Finally, server workload efficiency, a key factor influencing water use per digital service, is often overlooked by sustainability practitioners. Workload efficiency is affected by variables such as server utilization, power efficiency, refresh rates, and inactive server percentages ((Masanet et el., 2020) Shehabi et al., 2016). Addressing these factors is essential for a comprehensive understanding of data center water use per workload, which should account for both WUE-site and WUE-source determinants alongside the factors influencing server workload efficiency.

In this study, we aim to fill above knowledge gaps through a comprehensive analysis of overall data center water use per workload (Fig. 1). To make a fair comparison and gain a deeper understanding of the water use of data centers at the workload hosting level, we integrated the workload throughput score (i.e., *ssj_ops*) from the SPECpower_ssj 2008 benchmark into our analysis (Standard Performance Evaluation Corporation, 2022). This allowed us to create a new metric: the overall data center water use per workload, expressed in liters per workload ($\frac{liters}{ssj_ops}$), which addresses the limitations of the WUE-site and WUE-source metrics (Horner and Azevedo, 2016). We follow this with a decomposition analysis of the defined overall data center water use per workload, analyzing the determinants of water use and their

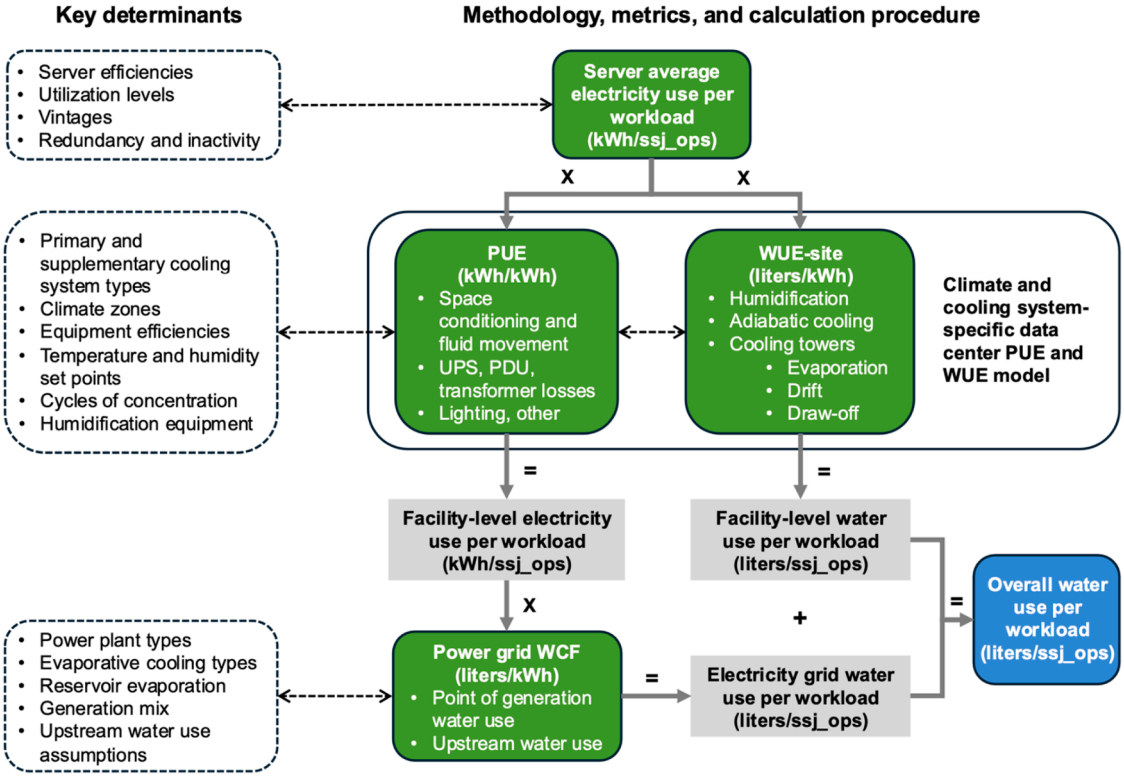



**Fig. 1.** Schematic of the overall water use per workload analysis approach and key systems determinants affecting the results. Determinants are depicted in dashed boxes linked to the variables (in green boxes) that they influence. The basic calculation procedure is indicated by grey arrows, with intermediate results highlighted in grey boxes.

interrelationships through the examination of real-world situations. In the decomposition analysis, we examined three key aspects of data center water use per workload in step-by-step fashion to make the influence of key determinants clear: (1) the relationship between server workload efficiency and factors such as server utilization level, power efficiency, refresh cycles, and inactive server percentage; (2) the effect of cooling technology, climate, and infrastructure efficiency on PUE, WUE-site, and their relationships to each other as well as how these interact with server workload efficiency to further impact the relationship between facility electricity and water use per workload; and (3) the determinants of WUE-source and how they interact with server workload efficiency to further influence the overall data center water use per workload.

Our analysis confirmed that considering workload efficiency is crucial in comparing water use between data centers, and the measures identified in our analysis can be effective for optimizing data center water use per computing workload from different perspectives. Additionally, our work can be readily extended to other measures of data center workloads derived from different testing methods when sufficient data exist, such as MLPerf for artificial intelligence workloads (MLCommons 2024), using the framework we present. This is an important step towards achieving improved water use efficiency in data centers.

## 2. Methods

### 2.1. Data center water use

The scope of this water use analysis includes both direct and indirect water consumption by data centers per workload. Direct water use encompasses water consumed at the data center for space humidification, direct adiabatic cooling of air, and/or heat rejection in cooling towers (including evaporated water, windage loss, and draw-off water). Indirect water use refers to water consumed through electricity generation, covering both upstream water use and water consumed at the point of power generation. Upstream water use includes water consumed in mining, extracting, cleaning, and processing of thermal power plant fuel inputs and in the manufacturing of renewable power equipment, such as solar photovoltaic (PV) panels and wind turbines. Point of generation water use refers to water consumed at the power plant itself due to the use of cooling towers, evaporation in cooling ponds and hydroelectric reservoirs, and periodic cleaning of solar PV panels.

Our decomposition analysis utilizes the equations outlined below, with the corresponding inputs and their sources summarized in Section S10 of the supporting information (SI). The overall data center water use per workload ($\frac{liters}{ssj_ops}$) is defined as Eq. (1).

$$W_{overall} = W_{site} + E_{site} \times WCF \tag{1}$$

where $W_{site}$ is the facility-level water use per workload ($\frac{liters}{ssj_ops}$, Eq. (2)), $E_{site}$ is the facility-level electricity use per workload ($\frac{kWh}{ssj_ops}$, Eq. (3)), and $WCF$ is the electrical grid water consumption factor, defined as the amount of freshwater consumed (in liters) per kilowatt-hour ($\frac{liters}{kWh}$) of grid electricity generated (Lee et al., 2018).

$$W_{site} = WUE_{site} \times e_{server} \tag{2}$$

where $WUE_{site}$ is the water usage effectiveness at the data center facility level ($\frac{liters}{kWh}$), $e_{server}$ is the average server electricity use per workload (i.e., server workload efficiency) ($\frac{kWh}{ssj_ops}$, Eq. (5)).

$$E_{site} = PUE \times e_{server} \tag{3}$$

where $PUE$ is the power usage effectiveness, defined as the ratio of a data center's total electricity use to its IT equipment electricity use ($\frac{kWh}{kWh}$) (The Green Grid, 2011b).

Substituting Eq. (2) and (3) into Eq. (1) gives Eq. (4):

$$W_{overall} = (WUE_{site} + PUE \times WCF) \times e_{server} = WUE_{source} \times e_{server} \tag{4}$$

where $WUE_{source}$ is the source-based water usage effectiveness ($\frac{liters}{kWh}$) (The Green Grid 2011a).

### 2.2. Direct server electricity use per workload

We focused on servers as the main component of IT loads because they are responsible for workload hosting, which is the main digital service associated with most data centers. However, future work could explore additional metrics for data center services, such as AI training and inference performance c(Masanet et al., 2024), or installed storage capacity in storage focused data centers. The server electricity use per workload at the data center level was modeled as a function of average server utilization level, the vintage distribution of the installed server stock (or refresh cycle), average server efficiency, and the fraction of inactive servers (including comatose servers and redundant servers operating at idle state) within the installed server stock (Lei, 2022; N. Lei and Masanet, 2021; Shehabi et al., 2016), expressed as Eq. (5).

$$e_{server} = 3.6 \times \frac{\widetilde{P}}{\widetilde{Th}} = 3.6 \times \frac{\varphi\overline{P}_i + (1-\varphi)\overline{P}_a}{(1-\varphi)\overline{P}_a\overline{Perf}} \tag{5}$$

where $\widetilde{P}$ is the weighted-average power use of the servers ($W$), and $\widetilde{Th}$ is the weighted-average server throughput ($\frac{ssj_ops}{s}$). $\overline{P}_i$ is the average power use of idle servers ($W$, Eq. (6)), $\overline{P}_a$ is the average power use of active servers ($W$, Eq. (7)), $\overline{Perf}$ is the average server power-performance ($ssj_ops/W$, Eq. (8)), and $\varphi$ is the fraction of inactive servers in the stock.

$$\overline{P}_i = \sum_y \varnothing_y \times P(u=0, y) \tag{6}$$

$$\overline{P}_a = \sum_y \varnothing_y \times P(u, y) \tag{7}$$

$$\overline{Perf} = \sum_y \varnothing_y \times Perf(u, y) \tag{8}$$

where $\varnothing_y$ is the server vintage distribution, represented by the percentage of installed servers of year $y$ in the server stock. $P(\cdot)$ and $Perf(\cdot)$ are, respectively, the server power and power-performance model, which are formulated as functions of the server utilization level ($u$) and the server installed year ($y$) using the local linear kernel regression approach based on the SPECpower_ssj 2008 benchmark database (Standard Performance Evaluation Corporation, 2022). Due to variability in server power-performance each year $y$, we analyzed $P(\cdot)$ and $Perf(\cdot)$ at different power efficiency quantiles for the pool of servers reported by the SPECpower_ssj 2008 database. Further details are available in SI Sections S1 and S2.

### 2.3. Climate and technology-specific PUE and WUE-site

PUE and WUE-site values are two generally-accepted metrics to report the energy and water use of data center operations (The Green Grid, 2011b, 2011a). As demonstrated in previous studies, the PUE and WUE-site values of data centers are simultaneously influenced by cooling system technologies, climate conditions, and operating efficiency practices (Lei and Masanet, 2022, 2020). However, they are also sporadically reported by operators and are poorly understood by sustainability analysts (Mytton 2021). Given their importance and sometimes competing effects in determining overall data center water use (Eq. (4)), we conducted an in-depth review and simulations of PUE and WUE-site across different cooling systems and climate scenarios (SI Section S3). In these simulations, the PUE results account for all heat generation sources and power-consuming devices, including power used

by IT equipment, lighting, fans, pumps, and losses from power transmission and distribution (Lei and Masanet, 2020). Meanwhile, WUE-site results consider water consumption associated with direct adiabatic cooling, humidification systems, and cooling tower operations (where applicable) (Lei and Masanet, 2022). Specifically, cooling tower water use includes water consumed for heat removal (sometimes referred to as greywater (Aldaya et al., 2012) in its functional role), windage loss, and draw-off water losses necessary to control chloride or mineral concentrations. Our aim was to clarify the interrelationships between PUE and WUE-site values and further illustrate how their determinants affect overall data center water use in the current study. To thoroughly quantify the impact of the various determinants, we covered typical data center cooling technologies and operating conditions (Evans, 2004; Lei and Masanet, 2022, 2020), as well as major types of global climate zones specified by American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) (ASHRAE, 2013). In total, we considered ten cooling technologies including (1) air-cooled chillers with no economizer, (2) airside economizers using air-cooled chillers as supplemental cooling, (3) airside economizers using water-cooled chillers as supplement cooling, (4) airside economizers and direct adiabatic cooling using air-cooled chillers as supplemental cooling, (5) airside economizers and direct adiabatic cooling using water-cooled chillers as supplemental cooling, (6) direct expansion systems, (7) water-cooled chillers with no economizer, (8) waterside economizers using water-cooled chillers as supplemental cooling, (9) IT liquid cooling using waterside economizers and using water-cooled chillers as supplemental cooling, (10) IT liquid cooling using dry coolers (a type of air-cooled heat exchanger) supporting adiabatic assist and using air-cooled chillers as supplemental cooling. Note that in these cases, supplemental cooling refers to cooling systems that are activated only when the primary cooling method (e.g., economizers, dry coolers, or adiabatic cooling) cannot maintain the desired temperature setpoints, which typically occurs during extreme weather conditions when free cooling methods alone cannot adequately dissipate heat. Additionally, we considered data center operations under nineteen different climate zones including extremely hot humid (0A), extremely hot dry (0B), very hot humid (1A), very hot dry (1B), hot humid (2A), hot dry (2B), warm humid (3A), warm dry (3B), warm marine (3C), mixed humid (4A), mixed dry (4B), mixed marine (4C), cool humid (5A), cool dry (5B), cool marine (5C), cool humid (6A), cold dry (6B), very cold (7), subarctic/arctic (8). Collectively, our combinations of cooling technologies and climate zones provide comprehensive global coverage. Finally, we took different data center operating efficiency practices, which were documented in previous literature, into consideration (Lei and Masanet, 2022; N. 2021, 2020). These different efficiency practices are represented by different quantiles of PUE and WUE-site simulation values for data centers implementing a given cooling technology in a given climate zone. Further details are provided in Section S3 of the SI.

### 2.4. Electric power water consumption factors

To thoroughly explore the indirect water use of data centers and to quantify its contribution to overall water use per workload, we analyzed water consumption factors (WCFs) across various electricity generation sources. Specifically, we assessed how different power sources—including geothermal, hydro, biomass, oil, coal, natural gas, solar photovoltaics (PV), and wind—affect overall data center water use per workload. This analysis covers both upstream water uses (e.g., water used in fuel extraction, processing, and transportation), and water consumed during power generation (Peer et al., 2019 M.A.B. Siddik et al., 2021), offering a more comprehensive assessment of water usage across the electricity generation life cycle (SI Section S4). However, it should be noted that the WCFs used in this study are based on generation-weighted U.S. averages. Since WCFs can vary significantly by region, power plant cooling technology, prime mover technology, and allocation methods (Grubert and Sanders, 2018; Lee et al., 2018), future research could investigate the impact of these factors to better align with the goal of accurately assessing their contribution to overall data center water use in non-U.S. regions.

## 3. Results

### 3.1. Determinants of direct server electricity use per workload

Our decomposition analysis starts at analyzing the factors that contribute to the server electricity use per data center workload. Fig. 2 shows the impact of various factors on server electricity use per workload, obtained using Eqs. (5) ~ (8) and assuming that the newest servers in the data center were purchased in 2023. The result shows that the lowest electricity use per workload is generally associated with efficient servers, high server utilization levels, fast refresh cycles, and low fractions of inactive servers. However, none of these factors alone is usually sufficient to justify low direct electricity use per workload, as indicated by how different combinations of these factors result in the same server electricity per workload (i.e., the same color) in Fig. 2. For example, a high-efficiency server pool (95th percentile) running at a low average utilization level and with many inactive servers can consume the same electricity per workload as a low-efficiency server pool (5th percentile) running at high utilization levels with few inactive servers (see points A and B, respectively). Server electricity use per workload is mainly affected by server efficiency. Efficient servers consume less power at idle state, and typically have larger dynamic ranges and better power proportionality (Barroso and Hölzle, 2007; Ryckbosch et al., 2011), leading to lower electricity use per workload (SI Fig. S5.1). Increasing server utilization level is another strategy to reduce direct electricity use per workload in data centers; however, the benefits diminish once server utilization exceeds around 40 % (SI Fig. S5.2). Optimizing the server refresh cycle reduces electricity use per workload, and at all utilization levels (SI Fig. S5.3), in a manner similar to using power efficient servers because servers become more power efficient over time. Likewise, removing inactive servers from the data center can reduce total electricity use per workload (SI Fig. S5.4), since idle servers consume power but are not performing any computing tasks. However, newer generations of servers also have significantly lower idle power, diminishing the effect of inactive servers over time.

### 3.2. Determinants of facility-level water use per server workload

Next in our decomposition analysis, we analyzed facility-level water use per workload as expressed in Eq. (2). Fig. 3 shows the assessed relationships between WUE-site, server electricity use per workload, and facility water use per workload. For a given server electricity use per workload, facility water use per workload increases with the increase of WUE-site, whereas server electricity use per workload determines the rate of this increase. Notably, low facility water use per workload is not necessarily a result of a low WUE-site value. This situation is indicated by points A and B in Fig. 3, in which both points have the same facility water use per workload but have vastly different values of WUE-site and server electricity use per workload. The interrelationships highlighted in Fig. 3 demonstrate the necessity of considering both WUE-site and server workload efficiency when considering the water efficiency of a given data center, as opposed to WUE-site alone.

### 3.3. Relationships between PUE and WUE-site values

Fig. 4. shows the interrelationships between PUE and WUE-site values clustered by ten different cooling technologies. The colored regions depicted for each cooling technology represent the range of variations in the relationship between PUE and WUE-site attributable to the combined effects of climate and data center infrastructure operating efficiency (see SI Fig. S6.1, which depicts how a region was defined for one cooling technology). It can be clearly observed that PUE and WUE-

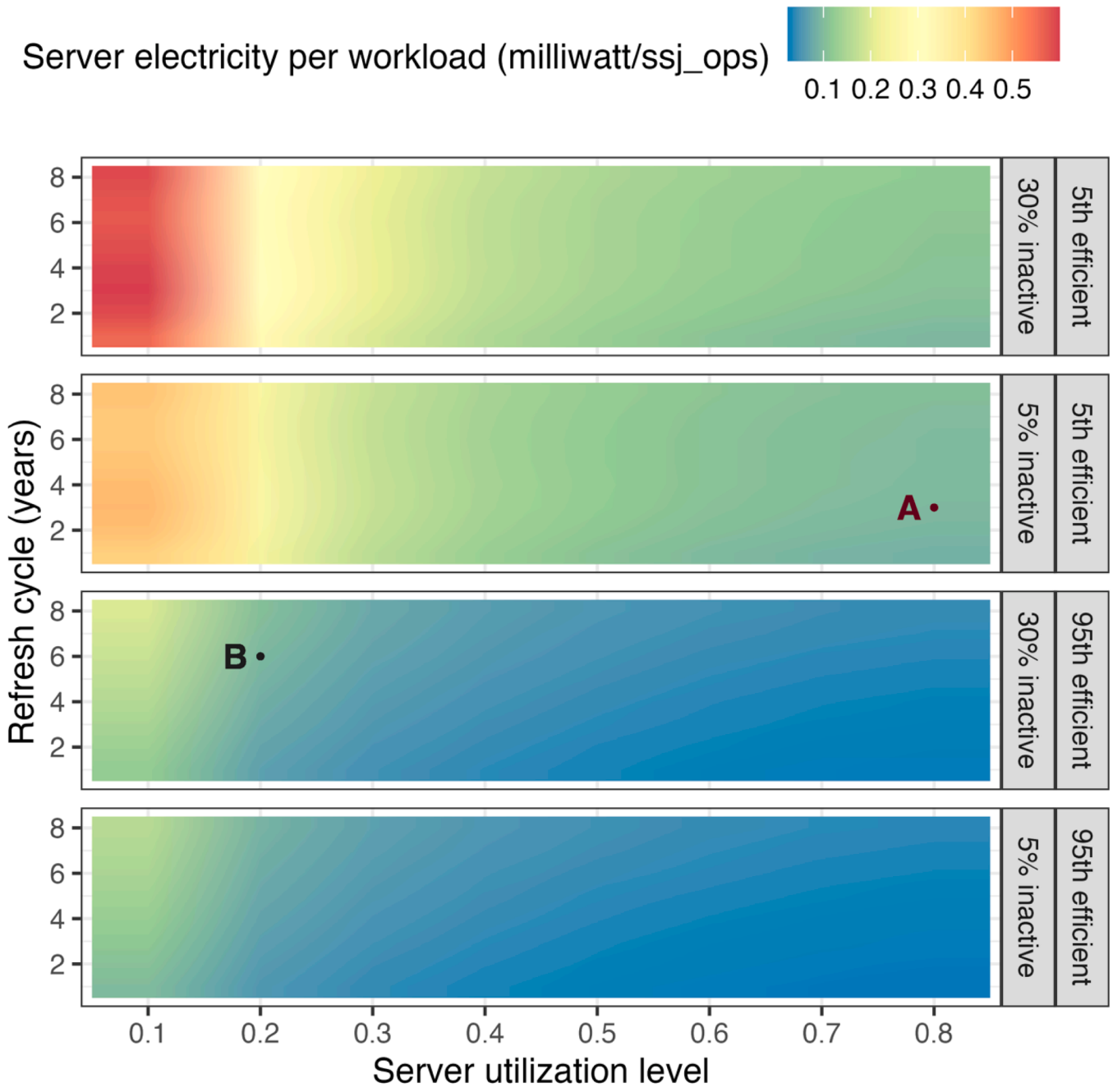



**Fig. 2.** Determinants of server electricity use per workload.
Notes: 5 % inactive indicates 5 % percent of the server pool is inactive servers; 30 % inactive indicates 30 % percent of the server pool is inactive servers; 5th efficient indicates the lowest 5 % of server power efficiency, i.e., the 5th percentile; 95th efficient indicates the highest 5 % of server power efficiency, i.e., the 95th percentile.

site values vary significantly by cooling technologies. Data centers that employ water-cooled chillers (as primary or supplemental cooling) have relatively higher WUE-site values due to large amounts of water consumed by cooling towers, particularly from evaporation. Conversely, data centers using direct-expansion systems or air-cooled chillers (as primary or supplemental cooling) have relatively lower WUE-site values, but typically have higher PUE values due to lower chiller coefficient of performance (COP) values (Lei and Masanet, 2022). Additionally, data centers that employ liquid cooling coupled with dry coolers, economizers, or adiabatic cooling systems can reduce both PUE and WUE-site values simultaneously, with the greatest electricity and water savings achieved for data centers using airside economizers and direct adiabatic cooling. Notably, data centers adopting IT liquid cooling systems can further reduce their PUE and WUE-site values owing to enhanced heat transfer efficiency, elevated coolant/facility water temperature setpoints, and the potential reduction or elimination of fan and chiller power (ASHRAE, 2014; Iyengar et al., 2012). Within each cooling technology cluster, we note a non-linear and positive relationship between PUE and WUE-site, where favorable climates and high operational efficiencies result in low PUE and WUE-site values, and hot and/or humid climates and low operational efficiencies lead to high PUE and WUE-site values (SI Fig. S6.1). Finally, we note that similar PUE and WUE-site values can be achieved with different combinations of cooling technology, climate, and efficiency, represented by overlaps between clusters. For example, points A and B have the same WUE-site value but PUE values differ by 2.0, while points C and D have the same PUE value but WUE-site values differ by 2.6. This situation again highlights the importance of system-wide optimization that simultaneously considers

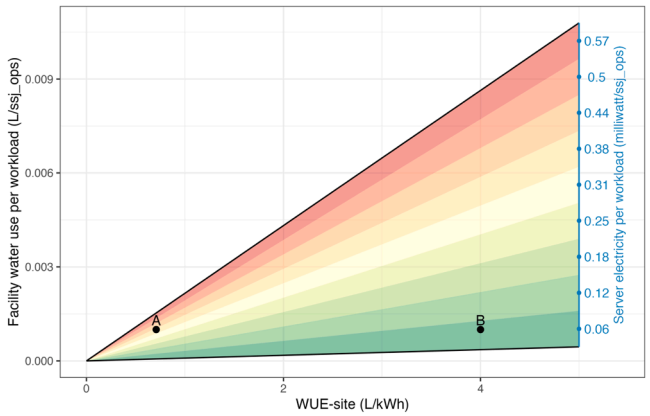



**Fig. 3.** The relationship between WUE-site, server electricity per workload, and facility-level water use per workload.
Note: the server electricity per workload axis values were uniformly sampled between the minimum and the maximum value in Fig. 2.

technology, climate, and efficiency in PUE and WUE-site reduction.

### 3.4. Relationships between facility-level electricity use and water use per workload

As shown in Fig. 3, facility-level water use per workload is driven by WUE-site and average server workload efficiency. Combining the regions in Figs. 2 and 3 (via Eqs. (2) and (3)) yields Fig. 5, which plots the relationship between facility-level water use and facility-level electricity use on a per-workload basis. In Fig. 5, large regions of possible facility-level electricity and water performance emerge due the combined variations associated with climate, operational efficiencies, and server workload efficiencies.

Our analysis showed that the lowest data center facility-level electricity and water use per workload are generally associated with the highest server workload efficiencies (SI Fig. S7.1), but the achievable minima (bottom left points) vary significantly by cooling system archetype. The results also indicate a tradeoff between facility-level electricity and water use per workload when selecting data center cooling technologies. That is, data centers with air-cooled chillers and direct expansion systems typically use less water but more electricity per workload, whereas data centers using water-cooled chillers consume less electricity but more water per workload. The implementation of

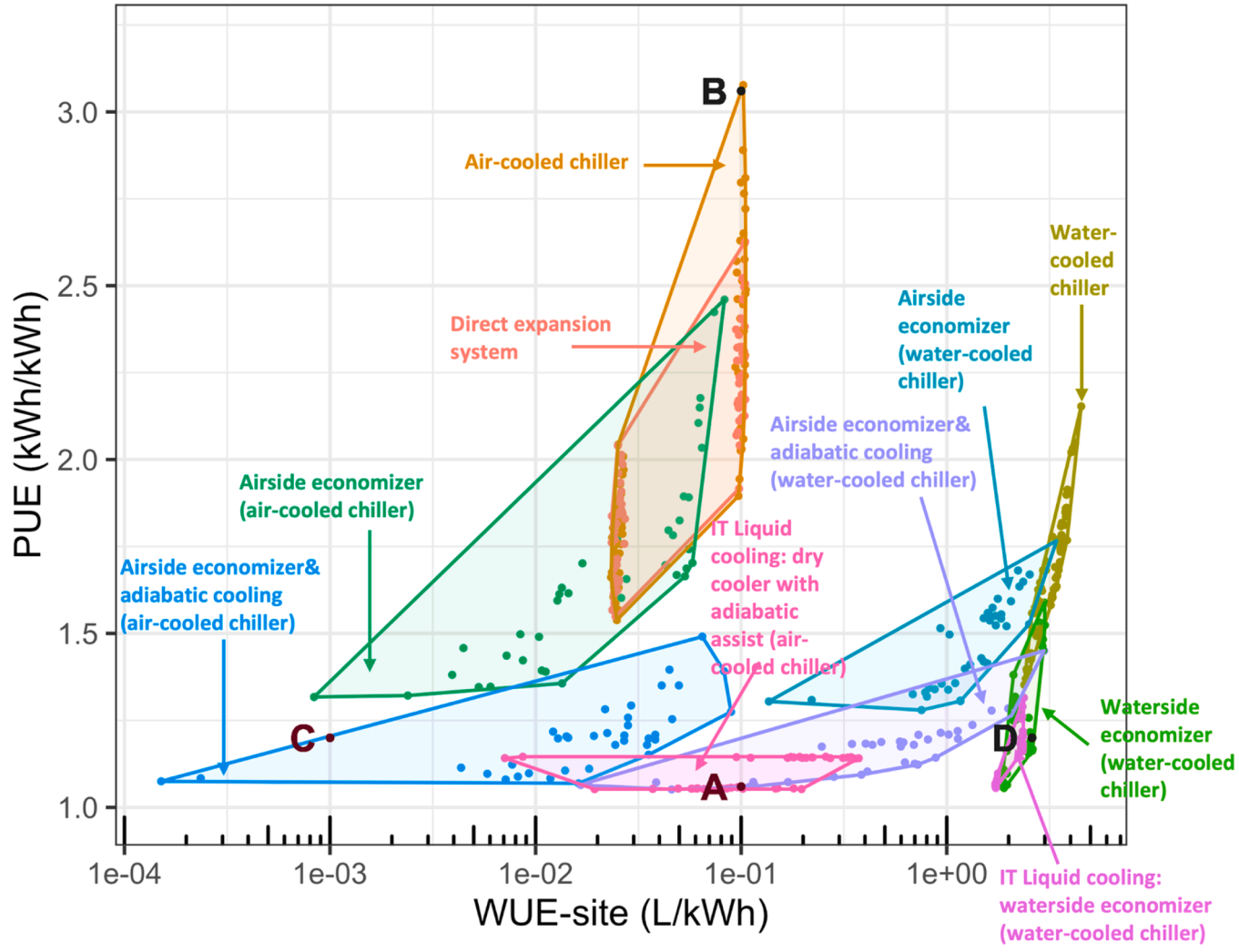



**Fig. 4.** Relationship between PUE, WUE-site, and cooling technology (x-axis values are logarithmically spaced).
Notes: See SI. Fig. S6.2 (where the x-axis is linearly spaced); Scatters within a given cooling technology consist of PUE and WUE-site values of different climate zones and data center operating efficiency practices (SI. Fig. S6.1); air-/water-cooled chillers in parentheses stand for supplemental cooling sources; PUE and WUE-site are yearly average values; IT liquid cooling includes scenarios of rear door heat exchangers, cold-plates, and immersion cooling (SI. Fig. S6.3).

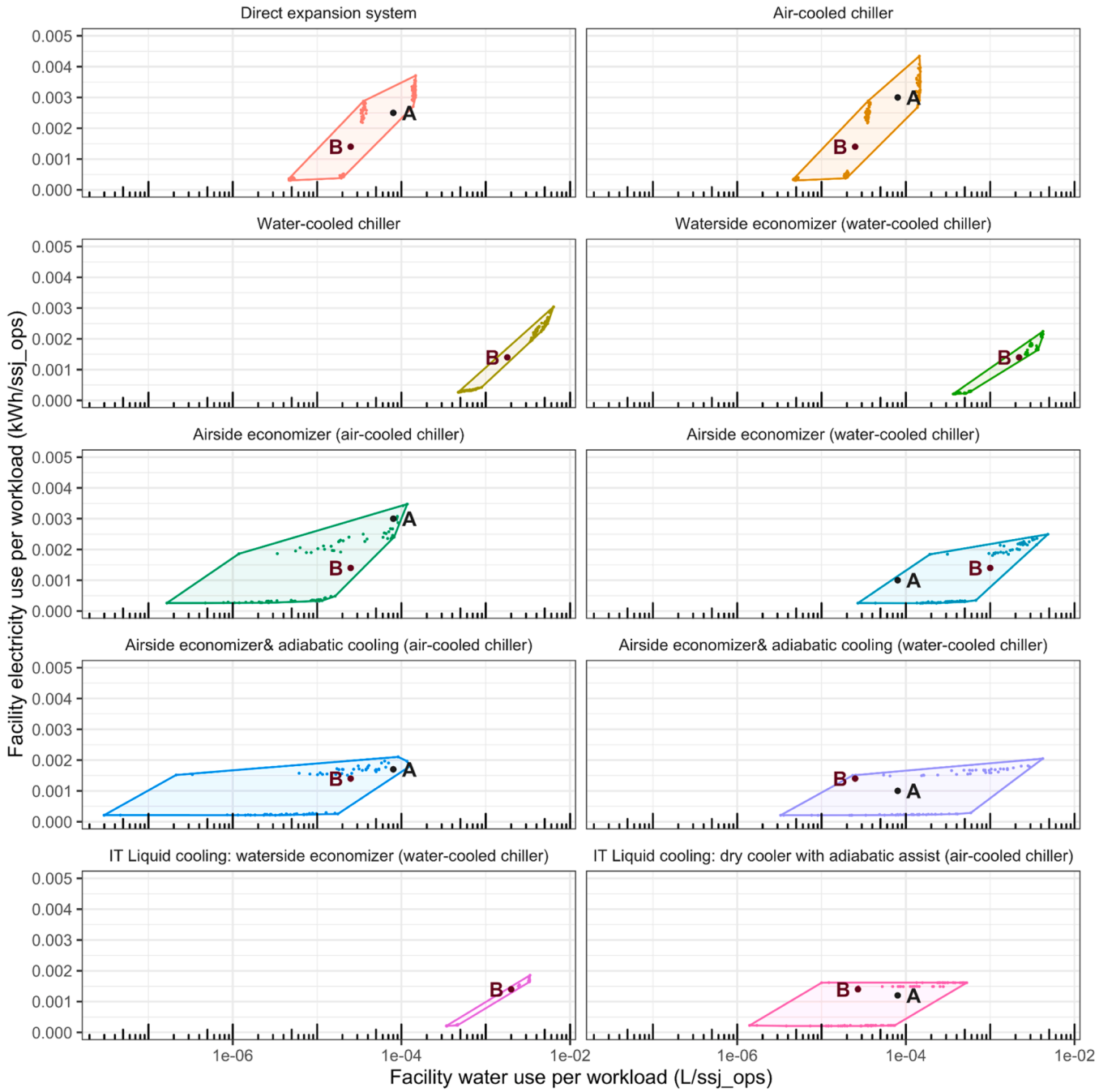



**Fig. 5.** Relationship between facility electricity and water use per workload by data center cooling system type.
Note: Points A have the same facility water use per workload; all points B have the same facility electricity use per workload.

economizers, dry coolers, liquid cooling, or adiabatic cooling in data centers can simultaneously reduce facility-level electricity and water use per workload. Adiabatic cooling, in particular, has been found to significantly reduce electricity use per workload in these settings. However, different combinations of determinants can lead to the same electricity and/or water use per workload. For example, the points marked A have the same facility-level water use per workload, and the points marked B have the same facility-level electricity use per workload), demonstrating again that electricity and water use per workload cannot be simply assessed or compared using a single determinant. In contrast, these results emphasize that there are multiple paths to reducing facility-level electricity and/or water use per workload in data centers.

### 3.5. Determinants of WUE-source

WUE-source (a component of Eq. (4)) is often used as a proxy to quantify the overall water use efficiency (i.e., total water consumption in liters per kWh of electricity consumed) of a data center (M.A.B. Siddik et al., 2021; The Green Grid 2011a). To clearly illustrate the determinants of WUE-source, we first compared the relationship between WUE-source and WUE-site for two common cooling systems (i.e., airside economizer and adiabatic cooling (water-cooled chiller) and direct expansion systems) and two power sources (i.e., wind and geothermal), which results in two regions in Fig. 6(a). These regions are bounded from left to right by the range of WUE-site values, where the region width corresponding to the airside economizer and adiabatic cooling (water-cooled chiller) is larger than that corresponding to the direct

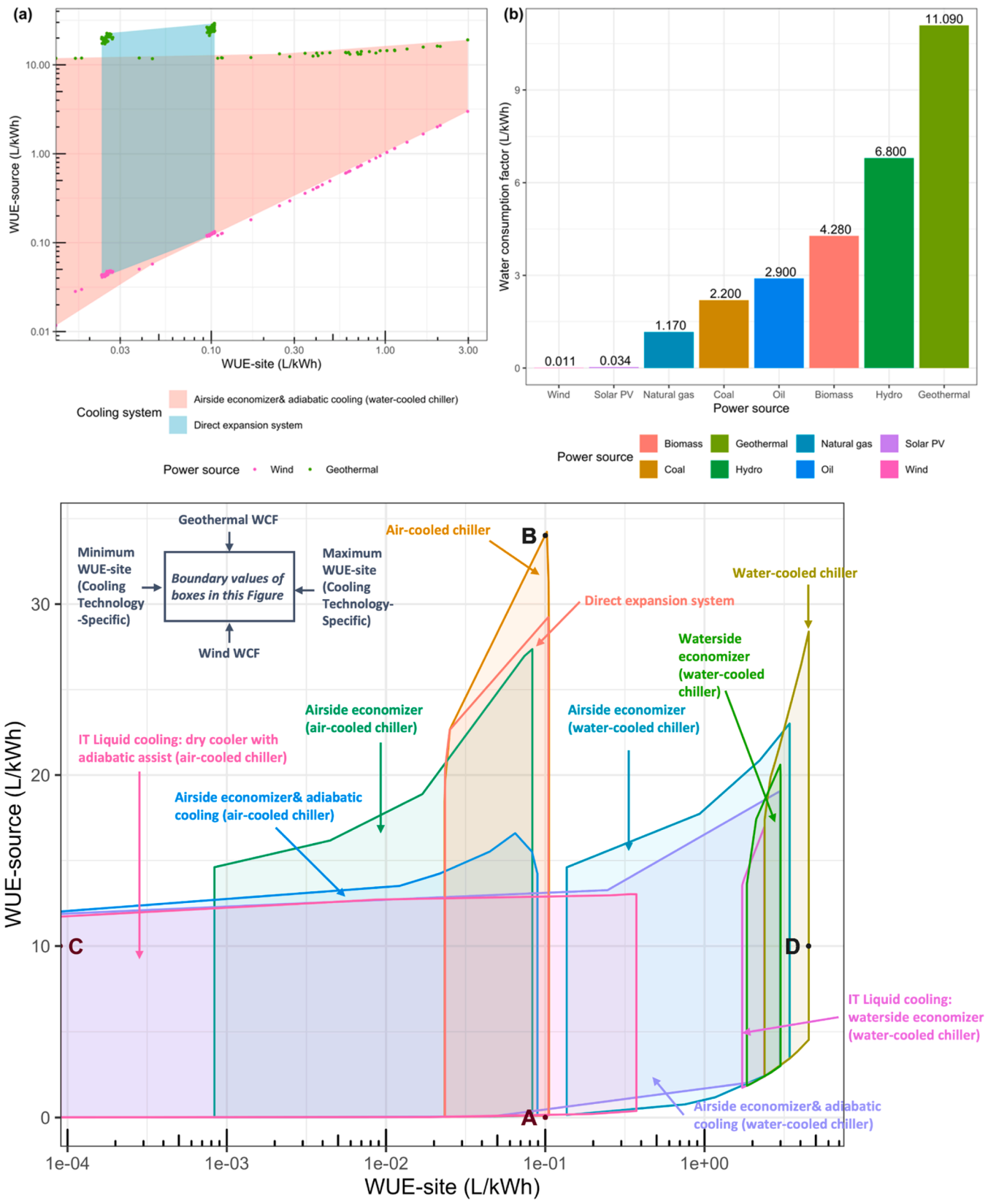



**Fig. 6.** (a) Relationship between WUE-source and WUE-site in data centers with airside economizer and adiabatic cooling (water-cooled chiller) or direct expansion systems (see SI Fig. S8.1, where the x-axis is linearly spaced); (b) U.S. water consumption factor by power source; (c) Relationship between WUE-source and WUE-site by cooling technology.

expansion system due to greater variability in WUE-site values (Fig. 4). For a given a cooling technology and a given WUE-site value, the distance between the lower and upper bound of the region is determined by the product of PUE value and WCF (see Eq. (4)), where the lower bound and upper bound is associated with the power source of the lowest U.S. WCF (wind) and the highest U.S. WCF (geothermal), respectively.

The WUE-source values of data centers are linearly correlated with both WUE-site values and the WCFs of power sources (Fig. 6(b)). Together, they define the range of potential WUE-source values for data centers (SI Fig. S8.1). Importantly, large gaps between the lower and upper boundaries of these regions (Fig. 6(a) and SI Fig. S8.1) highlights the huge influence of WCF in determining WUE-source. That is, data centers with high WUE-site values can achieve low WUE-source values by utilizing wind or solar power, while data centers with low WUE-site values can have high WUE-source values due to the use of fossil fuel, hydro, or geothermal based electricity.

Fig. 6(c) depicts the relationship between WUE-site and WUE-source for different data center cooling technologies, with lower and upper vertical bounds associated with minimum and maximal achievable WCFs (Fig. 6(b)), respectively. Due to differences in PUE and WUE-site values for different cooling technologies, the area and shape of the regions vary widely. For example, in Fig. 6(c), points A and B have the

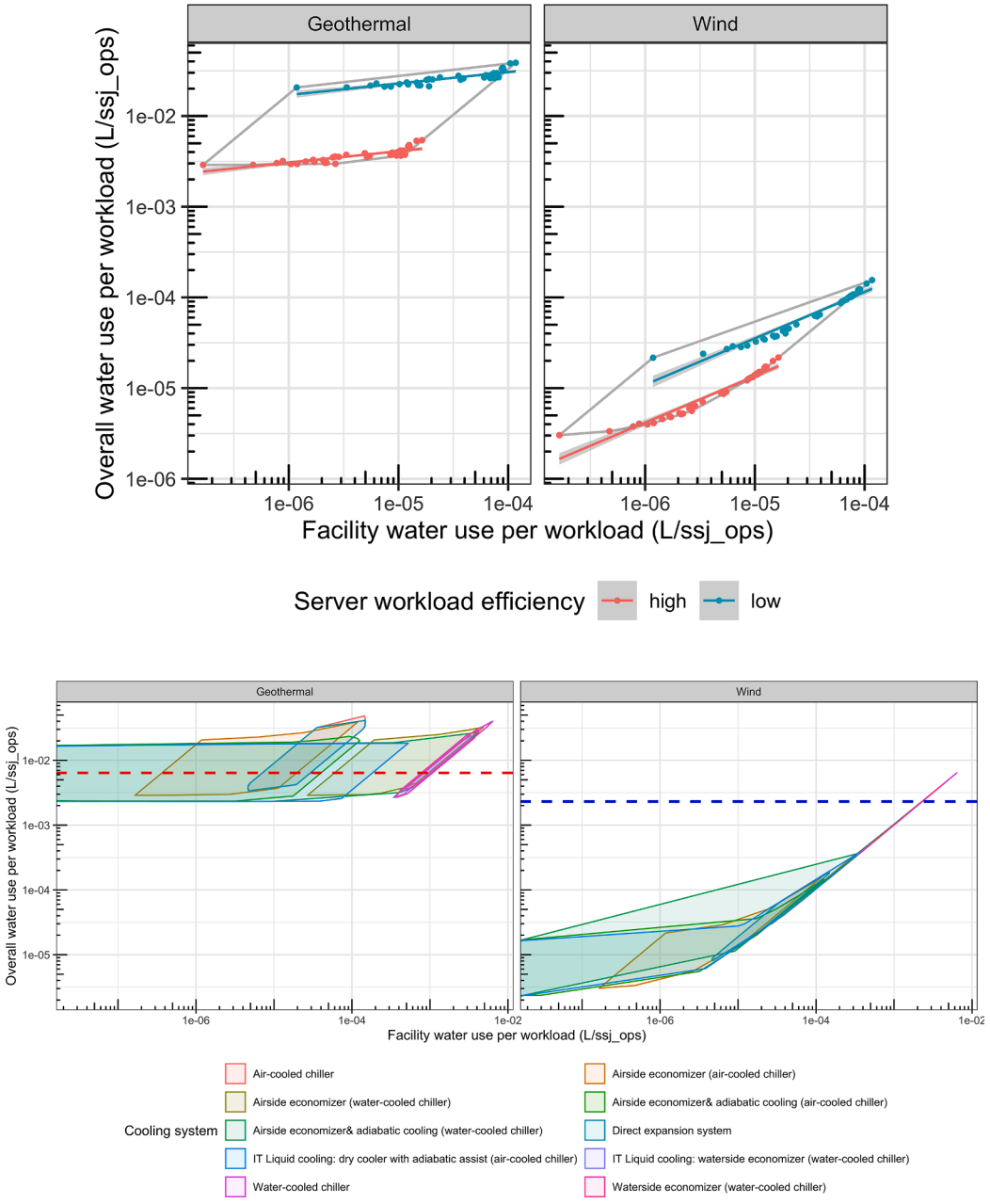



**Fig. 7.** (a): Relationship between data center facility-level and overall water use per workload for data centers with airside economizers (air-cooled chillers) and rely on geothermal or wind power.
Note: The regions were drawn by inputting the minimum and maximum server electricity per workload values from Fig. 2, along with the PUE and WUE-site values from Fig. 4 and the WCFs for geothermal and wind from Fig. 6(b), into Eq. (4).
Fig. 7(b): Relationship between data center facility-level and overall water use per workload by cooling technology for data centers that rely on geothermal/wind power (see SI Fig. S9.1 for data centers that rely on other power sources). Blue dashed line ~ minimum overall water use per workload for data centers using geothermal power; Red dashed line ~ maximum overall water use per workload for data centers using wind power.

same WUE-site but WUE-source values differ by a factor of 34, while points C and D have the same WUE-source but WUE-site values differ by a factor of 4.5. Depending on the product of the PUE and WCF values, data centers with the same WUE-site value can have very different WUE-source values. The same WUE-source value can be achieved by combining various determinants, which highlights two main approaches for data center WUE-source minimization to address local water issues and water conservation concerns. First, WUE-source can be reduced by improving data center operational efficiency, which can reduce both PUE and WUE-site, but the level of this reduction is limited by cooling technology implementation and data center climate conditions. Second, WUE-source can be reduced through choice of power sources with low WCFs, which is a more general approach that is not constrained by the cooling technology implementation, but rather the availability of electric grid options.

### 3.6. Determinants of overall water use per workload

The final step in our decomposition analysis was to assess, via Eq. (4), the interrelationships discussed in previous sections to quantify overall water use per workload. For simplicity, we analyzed data centers that rely on wind or geothermal power, corresponding to the power sources with the lowest and highest U.S. WCF in Fig. 6(b) respectively. Fig. 7(a) shows the relationship between overall and facility-level water use per workload for data centers using airside economizers and using air-cooled chillers as supplemental cooling equipment. Notably, the upper and lower bounds of the regions in Fig. 7(a) respectively correspond to low and high server workload efficiencies, and the relative distance between them represent the relative importance of server workload efficiency to the overall water use per workload in data centers. Furthermore, the relative distance of the lower and upper bound in Fig. 7(a) is larger for data centers that use geothermal power than those use wind power, indicating the significance of reducing overall water use per workload by improving server workload efficiency when a data center is constrained by power sources with high WCFs. Conversely, the relative distance between the lower and upper bound in Fig. 7(a) is small for data centers that use wind power, highlighting the importance of reducing overall water use by reducing facility-level water use per workload in data centers that rely on low WCF power sources.

Fig. 7(b) further shows the relationship between the facility-level and the overall water use per workload for data centers with different cooling systems. Data centers with low WCF power typically have a much lower overall water use than those using high WCF power, unless the data center has a significant on-site water consumption (e.g., the data center relies on water-cooled chillers or waterside economizers for cooling), which reconfirms the effectiveness of using low WCF electricity to reduce the water use of data centers. Additionally, the facility-level and overall water use per workload become more positively correlated as data centers transition from high to low WCF power, suggesting that more action should be taken to reduce facility-level water use (either through cooling system selection, PUE or WUE-site reduction, or server workload efficiency optimization) when low-WCF power is already available. Nevertheless, it is again clear that different combinations of determinants (WCF, cooling system type, and efficiency, etc.) can lead to similar overall water use per workload values, so data centers should not be compared on the basis of WUE-source, WCF, or WUE-site alone. This finding is again illustrated via the region between the two dashed lines in Fig. 7(b).

### 3.7. Sensitivity analysis

Fig. 8. shows the sensitivity results based on Sobol's total effect index (Sobol, 2001), which ranks the relative importance of the determinants of overall water use per workload via variance decomposition. Server efficiency quantile and server utilization level are the first and third largest contributors to variance in the overall water use result, as they

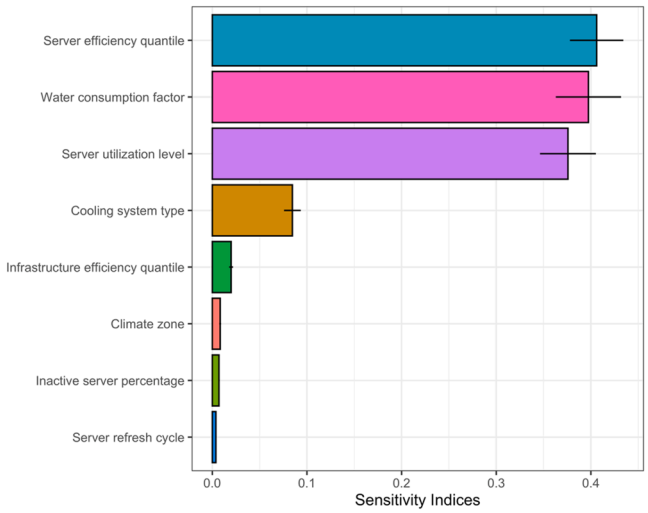



**Fig. 8.** Importance ranking (based on Sobol's total-effect index) for all of the determinants of the overall water use per workload (Error bar ~ 95 % confidence interval of sensitivity indices).
Note: Server efficiency quantile ~ measures the server power efficiency for a given year (SI S2); infrastructure efficiency quantile ~ measures the operating efficiency of the data center infrastructure (i.e., PUE and WUE-site) for a given cooling technology under a given climate zone.

have large impact on direct server electricity use per workload, which signifies that overall data center water use per workload can also be reduced through installing power-efficient servers (e.g., Energy Star certified servers) or server workload management (e.g., consolidation and virtualization). The WCF is the second largest contributor to the variance in the overall water use per workload results, suggesting that powering the data center with low-WCF electricity such as wind or solar PV can be effective in reducing overall data center water use. Cooling system type, data center infrastructure operating efficiency, and climatic conditions are important factors in determining data center PUE and WUE-site, making them non-trivial factors in data center overall water use minimization. Inactive server percentage and server fresh cycle also affect the overall water use per workload through affecting server workload efficiency, further confirming the usefulness of server management through removal of needless equipment or equipment refurbishment.

## 4. Conclusion

Increasing digitalization and freshwater restrictions have brought data center water conservation into the spotlight. While more data center operators are beginning to track and report WUE-site and WUE-source, these metrics fall short in measuring water use at the workload level, which is the ideal basis for understanding and comparing water use in data centers.

This study focuses on analyzing the determinants of overall workload-level water use, enabling more direct comparisons across data centers with diverse site-specific characteristics. Using statistical analysis of recent server performance data, we first highlighted the relationship between server electricity efficiency, utilization levels, refresh rates, and inactive server management as determinants of direct server electricity use per workload. Next, we examined the relationship between PUE and WUE-site through climate- and technology-specific simulations of major data center cooling archetypes, analyzing their influence on facility-level electricity and water use per workload. Finally, we assessed the relationship between electrical WCFs, WUE-source, and overall water use per workload. In general, this paper addresses critical gaps in existing research and contributes in the following key ways:

- Our study provides a comprehensive workload-level water use metric, integrating traditional efficiency measures such as PUE, WUE-site, WCF, and server workload efficiency. This metric enables consistent and comparable evaluations of data centers with varying site-specific characteristics, offering a path toward standardizing industry reporting, which is currently inconsistent in scope, quality, and granularity.
- Unlike prior studies that focus primarily on either direct or indirect water use, this paper provides a holistic analysis, covering direct water use such as space humidification, adiabatic cooling, and cooling tower operations, as well as indirect water use, including upstream water required for electricity fuel production, processing, and transportation, and water use at the point of electricity generation. This comprehensive approach lays the groundwork for developing integrated water-saving strategies.
- We conduct an extensive parametric analysis of workload-level water use, considering various determinants such as server efficiency, electrical grid water consumption factors, server utilization, cooling system type, infrastructure efficiency, climate zone, inactive server percentages, and server refresh cycles. This analysis provides actionable insights and tailored strategies for reducing water use under diverse data center conditions.
- Uniquely, this paper incorporates server workload efficiency and its determinants into the analysis of data center water use per workload. This provides a foundational framework to understand and improve water use for different digital services, supporting policymakers and operators as they increasingly focus on the direct and indirect water consumption of digital services.

## 5. Implications and limitations

Our findings can help policymakers design common measures to promote water conservation in data centers. For example, further promoting renewable energy (wind or solar PV), the use of the most efficient servers, and techniques such as virtualization for maintaining high levels of server utilization can effectively reduce water uses at the national level, while maximizing service outputs. Our findings also highlight that there is often no “silver bullet” for addressing the overall water use associated with all types of data centers, so strategies should be developed on a case-by-case basis. For instance, if the data center is geographically constrained with respect to its electric grid or climate, increasing average server workload efficiency is the most effective strategy to reduce water use. Furthermore, judicious selection of cooling technologies during the design phase of a data center has great potential to reduce a data center’s water use, which can lessen competition for local water resources from data center on-site water consumption in the meantime.

Our analysis also illustrates the value of evaluating water use at the workload hosting level in data centers, which brings to light the necessity of considering the computing power efficiency of IT equipment. This is an aspect that is frequently overlooked by sustainability analysts and policymakers, who have historically focused on PUE and WUE metrics. Our analysis enables impartial comparisons of electricity and water consumption among data centers by proposing several workload-based indicators, including direct server electricity use per workload, facility-level water use per workload, and overall water use per workload. For example, a cloud data center that utilizes airside economizers and adiabatic cooling, in addition to using water-cooled chillers for supplementary cooling, may consume more water on-site per kilowatt-hour of electricity used compared to a traditional data center with a direct-expansion system. However, such a comparison may result in a biased view that the traditional data center is more water-efficient by ignoring the number of digital services that can be delivered for the same amount of on-site and off-site water use. Our analysis fills the gap in the current understanding of water uses in data centers by providing examples of different scenarios that result in the same overall water use per workload.

While the primary focus of this study is the overall water use of data centers, the conservation strategies we suggest also have the potential to impact other aspects of data center operations. Data center workload management through virtualization allows multiple applications to run on a single server, potentially reducing the required IT capacity and saving initial investment costs through proper capacity planning. Improving the efficiency of IT equipment and infrastructure can reduce the electricity and water usage per data service, leading to cost savings (i.e., electric and water bills) in data center operations. Increasing energy efficiency and using renewable energy sources like wind or solar power can also reduce carbon emissions and contribute to climate change mitigation efforts, which are crucial in the current global warming context. To summarize, reducing the water use of data centers has many co-benefits such as reduced resource usage, reduced electric and water costs, and reduced carbon emissions. This provides strong evidence for policy incentives and operational measures to further improve water use per workload in data centers.

There are several limitations to consider in this analysis. Because of limited information on server computational efficiency, we rely on the SPECpower_ssj 2008 benchmark database, which may not fully reflect the distribution of server computational efficiency in all data center operations over time. There are concerns that even the least efficient servers in the SPECpower_ssj 2008 benchmark database may be more efficient than typical volume servers on the market. This could result in an underestimation of the upper bounds of water uses in our analysis. Our study primarily examines servers as the main component of IT loads, as they are the primary hosts for workloads. However, future research could consider a broader range of IT loads, including graphics processing units (GPUs), storage, and network loads to provide a more comprehensive assessment. Additionally, the WCFs analyzed in this study are based on generation-weighted U.S. averages for different power sources, without accounting for variations by region, cooling technology, prime mover technology, and other factors (Lee et al., 2018). It is also important to further emphasize that the workload-level water use analyzed here takes into account only water consumption, not water withdrawals, which are also important for understanding the potential effect that data centers can have on local water resources. We recommend that decision makers consider using electric power water withdrawal factors in combination with the WCFs we used to assess total site water use in the future. Finally, this paper focuses on dissecting and analyzing the determinants of workload-level water use in data centers, a metric sometimes referred to as the water footprint in data center sustainability research (Lei et al., 2023; Ristic et al., 2015; M.A.B. Siddik et al., 2021). For example, (Aldaya et al., 2012; Ansorge and Stejskalová, 2025) have underscored the importance of adhering to precise definitions of the water footprint to avoid confusion in critical water studies. As interest in data center water use and footprint research grows, establishing a clear and standardized definition of the data center water footprint, specifically tailored to this industry, is essential to guide future research and policy development.

## CRediT authorship contribution statement

**Nuoa Lei:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jun Lu:** Writing – review & editing, Visualization, Validation, Formal analysis, Data curation, Software. **Arman Shehabi:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition. **Eric Masanet:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.resconrec.2025.108310.

## Data availability

Data will be made available on request.

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